

Janitza Power Meter MAPS Template (Instrumentation)

The Janitza Power Meter MAPS template is designed to capture the data from a Janitza Power Meter and display the different values.

All MAPS templates provide the following:

- an associated PLC function block
- a set of SCADA tags (Adroit agents) linked to the provided signals
- an associated SCADA graphic with inbuilt faceplates

This MAPS template is a Janitza power meter, which provides no control over the power meter it only reads power meter data.

This document describes the Basic Janitza Power Meter MAPS Template. For details of its required PLC memory usage; number of PLC I/O and SCADA size (scan points) - see the **Resources** table below.

Janitza Power Meter Functional Philosophy

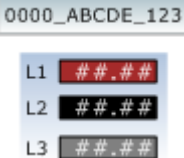
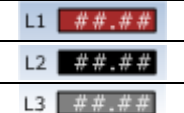
The power meter is connected via Modbus the block receives power signal which represents the various power measurements. These signals are then scaled according to the Current Transformer ratings from the Power Meter to display the various power measurements.

Field Interlocks

- None

Interlock fault recovery process – None

Typical Faceplate Graphics for the Janitza Power Meter MAPS Template

Janitza Power Meter				
				
	Flow Meter Control Name			
	L1 Current Display (Amps)			
	L2 Current Display (Amps)			
	L3 Current Display (Amps)			

Signal Description	Agent Type	Basic	
SCADA Control Words:			

Signal Description	Agent Type	Basic	
SCADA Status Words:			
SSSL 0 Voltage L1 - N	Analog	x	
SSSL 1 Voltage L2 - N	Analog	x	
SSSL 2 Voltage L3 - N	Analog	x	
SSSL 3 Voltage L1 - L2	Analog	x	
SSSL 4 Voltage L2 - L3	Analog	x	
SSSL 5 Voltage L2 - L1	Analog	x	
SSSL 6 Current in L1	Analog	x	
SSSL 7 Current in L2	Analog	x	
SSSL 8 Current in L3	Analog	x	
SSSL 9 Real power L1	Analog	x	
SSSL 10 Real power L2	Analog	x	
SSSL 11 Real power L3	Analog	x	
SSSL 12 Reactive power L1	Analog	x	
SSSL 13 Reactive power L2	Analog	x	
SSSL 14 Reactive power L3	Analog	x	
SSSL 15 Apparent power L1	Analog	x	
SSSL 16 Apparent power L2	Analog	x	
SSSL 17 Apparent power L3	Analog	x	
SSSL 18 Cosinus Phi in L1	Analog	x	
SSSL 19 Cosinus Phi in L2	Analog	x	
SSSL 20 Cosinus Phi in L3	Analog	x	
SSSL 21 THD U L1	Analog	x	
SSSL 22 THD U L2	Analog	x	
SSSL 23 THD U L3	Analog	x	
SSSL 24 THD I L1	Analog	x	
SSSL 25 THD I L2	Analog	x	
SSSL 26 THD I L3	Analog	x	
SSSL 27 Frequency L1	Analog	x	

SSSL 28	Cosinus Phi, Sum	Analog	x	
SSSL 29	Current in N	Analog	x	
SSSL 30	Real power, Sum	Analog	x	
SSSL 31	Reactive power, Sum	Analog	x	
SSSL 32	Apparent power, Sum	Analog	x	
SSSL 33	Mean value I in N	Analog	x	
SSSL 34	Mean value P, Sum	Analog	x	
SSSL 35	Mean value Q, Sum	Analog	x	
SSSL 36	Mean value S, Sum	Analog	x	
SSSL 37	Sum, real energy Wp, Consumption or HT1	Analog	x	
SSSL 38	Sum, real energy Wp, supply or NT2	Analog	x	
SSSL 39	Sum, reactive energy Wq, kap. or ind/HT3	Analog	x	
SSSL 40	Sum, reactive energy Wq, sum or ind/NT4	Analog	x	
SSSL 41	Sum, apparent energy	Analog	x	
SSSL 42	Mean values U L1-N	Analog	x	
SSSL 43	Mean values U L2-N	Analog	x	
SSSL 44	Mean values U L3-N	Analog	x	
SSSL 45	Mean values U L1-L2	Analog	x	
SSSL 46	Mean values U L2-L3	Analog	x	
SSSL 47	Mean values U L3-L1	Analog	x	

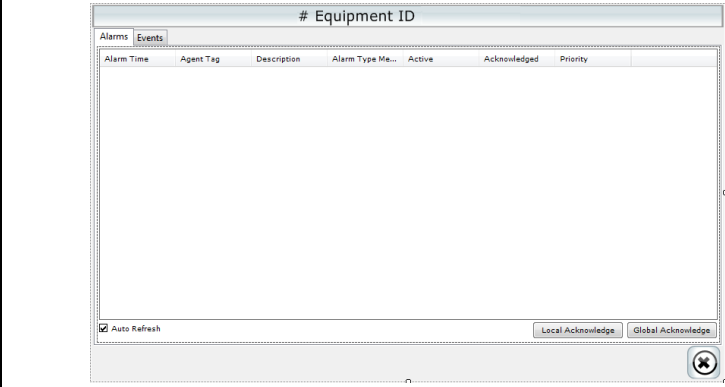
Digital Inputs:		Basic	
Digital Outputs:			
Analog Inputs:			
Analog Outputs:			
Resources		Basic	
Digital Inputs		0	
Digital Outputs		0	
Analog Inputs		0	
Analog Outputs		0	
PLC Memory Steps		465	
System Words Used		90	
System Bits Used		2	
SCADA SCAN Points		48	

Template: Advanced Flow Meter PLC Function Block

12_JPM_0001 (Janitza Power Meter Basic)		FB_12_JPM_0001	
		JPM_B_v1_0	
DATA_12_JPM_0001.VAR_From_PM200_220	From_PM200_220	S_SW1_V_L1_N	DATA_12_JPM_0001.S_SW1_V_L1_N
DATA_12_JPM_0001.VAR_From_PM269_297	From_PM269_297	S_SW2_V_L2_N	DATA_12_JPM_0001.S_SW2_V_L2_N
DATA_12_JPM_0001.VAR_From_PM422_437	From_PM422_437	S_SW3_V_L3_N	DATA_12_JPM_0001.S_SW3_V_L3_N
DATA_12_JPM_0001.VAR_From_PM600_603	From_PM600_603	S_SW4_V_L1_L2	DATA_12_JPM_0001.S_SW4_V_L1_L2
		S_SW5_V_L2_L3	DATA_12_JPM_0001.S_SW5_V_L2_L3
		S_SW6_V_L3_L1	DATA_12_JPM_0001.S_SW6_V_L3_L1
		S_SW7_C_L1	DATA_12_JPM_0001.S_SW7_C_L1
		S_SW8_C_L2	DATA_12_JPM_0001.S_SW8_C_L2
		S_SW9_C_L3	DATA_12_JPM_0001.S_SW9_C_L3
		S_SW10_P_L1	DATA_12_JPM_0001.S_SW10_P_L1
		S_SW11_P_L2	DATA_12_JPM_0001.S_SW11_P_L2
		S_SW12_P_L3	DATA_12_JPM_0001.S_SW12_P_L3
		S_SW13_ReacP_L1	DATA_12_JPM_0001.S_SW13_ReacP_L1
		S_SW14_ReacP_L2	DATA_12_JPM_0001.S_SW14_ReacP_L2
		S_SW15_ReacP_L3	DATA_12_JPM_0001.S_SW15_ReacP_L3
		S_SW16_AppP_L1	DATA_12_JPM_0001.S_SW16_AppP_L1
		S_SW17_AppP_L2	DATA_12_JPM_0001.S_SW17_AppP_L2
		S_SW18_AppP_L3	DATA_12_JPM_0001.S_SW18_AppP_L3
		S_SW19_CosPi_L1	DATA_12_JPM_0001.S_SW19_CosPi_L1
		S_SW20_CosPi_L2	DATA_12_JPM_0001.S_SW20_CosPi_L2
		S_SW21_CosPi_L3	DATA_12_JPM_0001.S_SW21_CosPi_L3
		S_SW22_THD_U_L1	DATA_12_JPM_0001.S_SW22_THD_U_L1
		S_SW23_THD_U_L2	DATA_12_JPM_0001.S_SW23_THD_U_L2
		S_SW24_THD_U_L3	DATA_12_JPM_0001.S_SW24_THD_U_L3
		S_SW25_THD_I_L1	DATA_12_JPM_0001.S_SW25_THD_I_L1
		S_SW26_THD_I_L2	DATA_12_JPM_0001.S_SW26_THD_I_L2
		S_SW27_THD_I_L3	DATA_12_JPM_0001.S_SW27_THD_I_L3
		S_SW28_F_L1	DATA_12_JPM_0001.S_SW28_F_L1
		S_SW29_CosPhi_Sum	DATA_12_JPM_0001.S_SW29_CosPhi_Sum
		S_SW30_C_N	DATA_12_JPM_0001.S_SW30_C_N
		S_SW31_RealP_Sum	DATA_12_JPM_0001.S_SW31_RealP_Sum
		S_SW32_ReacP_Sum	DATA_12_JPM_0001.S_SW32_ReacP_Sum
		S_SW33_AppP_Sum	DATA_12_JPM_0001.S_SW33_AppP_Sum
		S_SW34_Mean_I_N	DATA_12_JPM_0001.S_SW34_Mean_I_N
		S_SW35_Mean_P_Sum	DATA_12_JPM_0001.S_SW35_Mean_P_Sum
		S_SW36_Mean_Q_Sum	DATA_12_JPM_0001.S_SW36_Mean_Q_Sum
		S_SW37_Mean_S_Sum	DATA_12_JPM_0001.S_SW37_Mean_S_Sum
		S_SW38_Wp_Cons_Sum	DATA_12_JPM_0001.S_SW38_Wp_Cons_Sum
		S_SW39_Wp_Supply_Sum	DATA_12_JPM_0001.S_SW39_Wp_Supply_Sum
		S_SW40_Wq_Reactive_kap	DATA_12_JPM_0001.S_SW40_Wq_Reactive_kap
		S_SW41_Wq_Reactive_Sum	DATA_12_JPM_0001.S_SW41_Wq_Reactive_Sum
		S_SW42_Sum_App	DATA_12_JPM_0001.S_SW42_Sum_App
		S_SW43_Mean_U_L1_N	DATA_12_JPM_0001.S_SW43_Mean_U_L1_N
		S_SW44_Mean_U_L2_N	DATA_12_JPM_0001.S_SW44_Mean_U_L2_N
		S_SW45_Mean_U_L3_N	DATA_12_JPM_0001.S_SW45_Mean_U_L3_N
		S_SW46_Mean_U_L1_L2	DATA_12_JPM_0001.S_SW46_Mean_U_L1_L2
		S_SW47_Mean_U_L2_L3	DATA_12_JPM_0001.S_SW47_Mean_U_L2_L3
		S_SW48_Mean_U_L3_L1	DATA_12_JPM_0001.S_SW48_Mean_U_L3_L1

[illegible]

Template Graphic: Advanced Alarms & Events Screen



Control

Descriptions

Alarms

Display all the alarms filtered for the current Digital Output

Events

Display all the events filtered for the current Digital Output.

Local Acknowledge

Acknowledge all the alarms on locally on the current machine

Global Acknowledge

Acknowledge all the alarms on globally on all the machines.